

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0523

Bear Island 138 kV
April 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0523. The Interconnection Customer proposes a 91.6 MW solar photovoltaic facility interconnecting at the Bear Island 138 kV substation in the Virginia area. The request was received on October 15, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 238 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,168,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Contra Costa Substation-Birchwood 138.0 kV
Contingency: Nauvoo-Birchwood 345.0 kV
Loading: 129.6% (Rating: 360 MVA)
DFAX: 5.3%
Pre-project loading: 128.2%
2. Facility: El Dorado Refinery-Morrell 138.0 kV
Contingency: Bear Island-Dunn Siding 345.0 kV
Loading: 117.7% (Rating: 246 MVA)
DFAX: 6.4%
Pre-project loading: 115.3%
3. Facility: El Dorado Refinery-Lookout 138.0 kV
Contingency: Bear Island-Dunn Siding 345.0 kV Line
Loading: 115.2% (Rating: 239 MVA)
DFAX: 6.4%
Pre-project loading: 112.7%
4. Facility: Bear Island-Laporte Junction 138.0 kV L
Contingency: Dunn Siding-Bear Island 345.0 kV
Loading: 113.8% (Rating: 223 MVA)
DFAX: 17.7%
Pre-project loading: 106.5%
5. Facility: N Doswell-Bear Island 138.0 kV
Contingency: Bear Island-Dunn Siding 345.0 kV line
Loading: 112.5% (Rating: 225 MVA)
DFAX: 17.4%
Pre-project loading: 105.3%

6. Facility: Trenton Diesel-Laporte Junction 138.0 kV
 Contingency: Bear Island-Dunn Siding 345.0 kV
 Loading: 111.9% (Rating: 216 MVA)
 DFAX: 17.7%
 Pre-project loading: 104.4%
7. Facility: Trenton Diesel-North Doswell 138.0 kV
 Contingency: Bear Island-Dunn Siding 345.0 kV line
 Loading: 110.1% (Rating: 215 MVA)
 DFAX: 17.4%
 Pre-project loading: 102.6%
8. Facility: Lake Crk-Sierra Pacific Lumber 138.0 kV
 Contingency: Taber-Glenannan Substation 345.0 kV
 Loading: 109.0% (Rating: 997 MVA)
 DFAX: 5.5%
 Pre-project loading: 108.5%
9. Facility: Morrell-Trenton Diesel 138.0 kV
 Contingency: Tippecanoe-Skaggs Island Jct 2 138.0 kV
 Loading: 107.9% (Rating: 226 MVA)
 DFAX: 5.5%
 Pre-project loading: 105.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Contra Costa Substation-Birchwood 138.0 kV	Rebuild line with higher-capacity conductor	\$14,046,000	8% (\$1,199,000)
El Dorado Refinery-Morrell 138.0 kV	Rebuild line with higher-capacity conductor	\$7,158,000	25% (\$1,759,000)
El Dorado Refinery-Lookout 138.0 kV	Rebuild line with higher-capacity conductor	\$21,823,000	25% (\$5,362,000)
Bear Island-Laporte Junction 138.0 kV L	Rebuild line with higher-capacity conductor	\$7,756,000	26% (\$2,038,000)
N Doswell-Bear Island 138.0 kV	Rebuild line with higher-capacity conductor	\$8,229,000	26% (\$2,162,000)
Trenton Diesel-Laporte Junction 138.0 kV	Rebuild line with higher-capacity conductor	\$2,699,000	26% (\$709,000)
Trenton Diesel-North Doswell 138.0 kV	Rebuild line with higher-capacity conductor	\$2,433,000	26% (\$639,000)
Lake Creek-Sierra Pacific Lumber 138.0 kV	Reconductor line	\$884,000	1% (\$6,000)
Trenton Diesel-Morrel 138.0 kV L	Reconductor line	\$5,180,000	25% (\$1,294,000)
	Total allocated to project		\$15,168,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.